

Solving $Ax = 0$

The recipe for finding **every** do-nothing trade, even when you have more funds than rules.

Pivots, free variables, rank, special solutions.

Rank r

$n - r$ special solutions

MIT 18.06, Lecture 7

Four funds, three rules

Nifty, a Nifty ETF (its twin), bonds, balanced (half Nifty, half bonds). Rows: budget, return, exposure.

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix}$$

Real fund menus look like this: many products, a few things you measure, plenty of overlap. Which trades change **nothing**?

STEP 0

Elimination keeps the nullspace

Row operations only **recombine the rules**. A trade that leaves every rule at 0 leaves every combination of rules at 0 too.

So $Ax = 0$ and $Ux = 0$ have the **same** solutions.

CAREFUL

The column space **does** change. Use U for the nullspace, never for $C(A)$.

STEP 1

Eliminate to a staircase

$$U = \begin{bmatrix} \boxed{1} & 1 & 1 & 1 \\ 0 & 0 & \boxed{-5} & -2.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Column 2 has no pivot (the ETF copies Nifty), so skip it. Row 3 vanishes: exposure added no new information.

Pivot columns: Nifty, bonds. **Free columns:** ETF, balanced. **Rank** $r = 2$.

STEP 2

Pivot = forced, free = your choice

Choose the free amounts. The pivot amounts are then forced:

$$x_3 = -0.5 x_4 \quad x_1 = -x_2 - 0.5 x_4$$

RANK

r = number of pivots = number of genuinely different funds.

STEP 3

Special solutions

One free variable = 1, the others 0:

- ETF = 1: $s_1 = (-1, 1, 0, 0)$, swap Nifty for its twin
- Balanced = 1: $s_2 = (-0.5, 0, -0.5, 1)$, build balanced yourself

Check s_2 : $-0.5 - 0.5 + 1 = 0$, $-6 - 3.5 + 9.5 = 0$,
 $-0.5 + 0.5 = 0 \checkmark$

$N(A)$ = every mix $c s_1 + d s_2$: a plane in $\mathbb{R}^4$.

n – r special solutions

Funds	m	n	r	n – r
Nifty, bonds, midcap	3	3	3	0
Nifty, bonds, balanced	3	3	2	1
Nifty, ETF, bonds, balanced	3	4	2	2

THE RULE

More funds than rules ($n > m$) $\Rightarrow$ a do-nothing trade always exists.

R: clear above the pivots too

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 \\ 0 & 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot columns first: $R = [I \ F]$ with

$$F = \begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$$

The nullspace matrix

$$N = \begin{bmatrix} -F \\ I \end{bmatrix} = \begin{matrix} x_1 \\ x_3 \\ x_2 \\ x_4 \end{matrix} \begin{bmatrix} -1 & -0.5 \\ 0 & -0.5 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Its columns are the special solutions, and $RN = I(-F) + F I = 0$. No back substitution needed.

Four products, three scenarios

Futures, 20,000 call, 20,000 put, 19,900 call. The recipe finds one free column (the put) and one special solution:

$$(1, -1, 1, 0)$$

Long future, short call, long put pays 0 everywhere: **put-call parity**. The 19,900 call is in no do-nothing trade: it's the only one paying at 20,000.

Key takeaways

1. Row operations keep the nullspace.
2. Pivot variables are forced, free ones are chosen.
3. Rank r = number of pivots.
4. Special solution: one free = 1, others 0.
5. $n - r$ of them; $n > m \Rightarrow$ always one.
6. $R = [I \ F], N = \begin{bmatrix} -F \\ I \end{bmatrix}$.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A pivot and special-solution lab for any matrix up to 3×5 , and worked hedging and options examples:

learn.krisarj.com

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